

Link of Completion: <https://cyberdefenders.org/blueteam-ctf-challenges/achievements/sPo0f1ng/amadey-apt-c-36/>

Q1.In the memory dump analysis, determining the root of the malicious activity is essential for comprehending the extent of the intrusion. What is the name of the parent process that triggered this malicious behavior?

to determine the parent process tree here, i run the following volatility tools for this part and find the fake windows process that contain malware

`vol.py -q -f '/home/ubuntu/Desktop/Start here/Artifacts/Windows 7 x64-Snapshot4.vmem' windows.pstree`

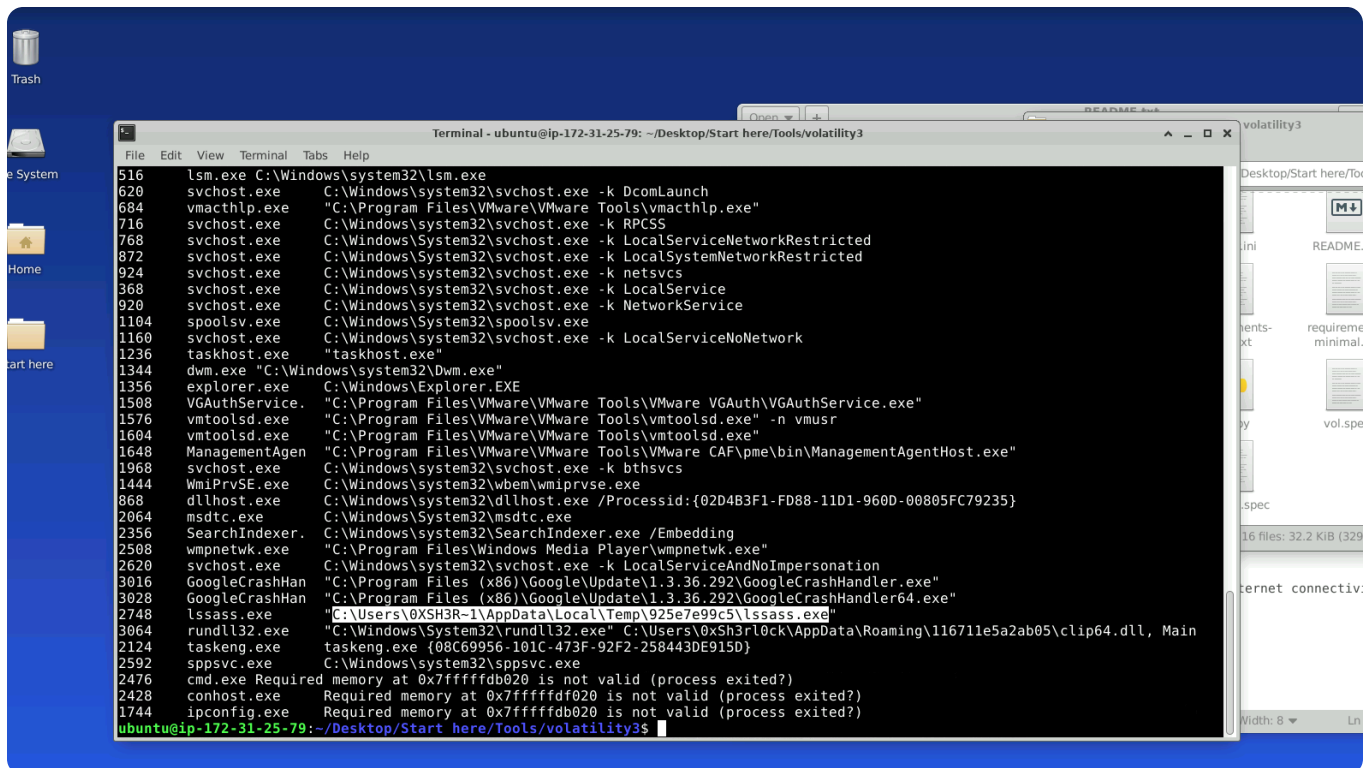
```
Terminal - ubuntu@ip-172-31-25-79: ~/Desktop/Start here/Tools/volatility3
File Edit View Terminal Tabs Help
** 920 500 svchost.exe 0xfa8002cc5100 15 396 0 False 2023-08-09
** 924 500 svchost.exe 0xfa8002b24b30 31 887 0 False 2023-08-09
*** 2124 924 taskeng.exe 0xfa80007f38e0 5 85 1 False 2023-08-09
** 2592 500 sppsvcs.exe 0xfa80007c8b30 4 144 0 False 2023-08-09
** 684 500 vmacthlp.exe 0xfa8002b7a6e0 3 54 0 False 2023-08-09
** 1968 500 svchost.exe 0xfa8001d41890 6 91 0 False 2023-08-09
** 2356 500 SearchIndexer.exe 0xfa8002f7d6f0 11 592 0 False 2023-08-09
** 2620 500 svchost.exe 0xfa80030c7b30 16 241 0 False 2023-08-09
** 1604 500 vmtoolsd.exe 0xfa800e9ff060 9 305 0 False 2023-08-09
*** 2476 1604 cmd.exe 0xfa80031591b0 0 - 0 False 2023-08-09
07.000000
**** 1744 2476 ipconfig.exe 0xfa8000784920 0 - 0 False 2023-08-09
9 21:50:07.000000
** 716 500 svchost.exe 0xfa8002b9f320 7 257 0 False 2023-08-09
** 2508 500 wmpnetwk.exe 0xfa800305cb30 10 208 0 False 2023-08-09
** 1104 500 spoolsv.exe 0xfa8002d12b30 13 266 0 False 2023-08-09
** 1236 500 taskhost.exe 0xfa8002dc6270 8 145 1 False 2023-08-09
** 1508 500 VGAuthService.exe 0xfa8002e8e940 3 88 0 False 2023-08-09
** 868 500 dlhst.exe 0xfa8001ef66d0 13 190 0 False 2023-08-09
** 872 500 svchost.exe 0xfa8002b12890 18 435 0 False 2023-08-09
*** 1344 872 dwm.exe 0xfa8002e00060 4 70 1 False 2023-08-09
** 620 500 svchost.exe 0xfa8002b41970 10 351 0 False 2023-08-09
*** 1444 620 WmiPrvSE.exe 0xfa8002c135d0 10 287 0 False 2023-08-09
** 368 500 svchost.exe 0xfa8002ca7060 17 675 0 False 2023-08-09
** 1648 500 ManagementAgent.exe 0xfa80015f9370 10 92 0 False 2023-08-09
* 508 400 lsass.exe 0xfa8002a77b30 6 555 0 False 2023-08-09
408 392 csrss.exe 0xfa8002a6e120 10 225 1 False 2023-08-09
464 392 winlogon.exe 0xfa8002ab5700 3 111 1 False 2023-08-09
1356 1312 explorer.exe 0xfa8002cf5b30 21 685 1 False 2023-08-09
* 1576 1356 vmtoolsd.exe 0xfa8002eb9630 6 183 1 False 2023-08-09
3016 2144 GoogleCrashHandler.exe 0xfa800143eb30 5 91 0 True 2023-08-09
3028 2144 GoogleCrashHandler.exe 0xfa8001bfe6f0 5 81 0 False 2023-08-09
2748 2524 lsass.exe 0xfa800300a750 7 254 1 True 2023-08-09
* 3064 2748 rundll32.exe 0xfa8003042b30 1 64 1 True 2023-08-09
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$
```

ANSWER: lsass.exe

Q2.Once the rogue process is identified, its exact location on the device can reveal more about its nature and source. Where is this process housed on the workstation?

for this part to get the answer, i simple just edited the volatility command into "windows.cmdline"

vol.py -q -f '/home/ubuntu/Desktop/Start here/Artifacts/Windows 7 x64-Snapshot4.vmem'
windows.cmdline



```
516 lsass.exe C:\Windows\system32\lsass.exe
620 svchost.exe C:\Windows\system32\svchost.exe -k DcomLaunch
684 vmacthlp.exe "C:\Program Files\VMware\VMware Tools\vmacthlp.exe"
716 svchost.exe C:\Windows\system32\svchost.exe -k RPCSS
768 svchost.exe C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted
872 svchost.exe C:\Windows\system32\svchost.exe -k LocalSystemNetworkRestricted
924 svchost.exe C:\Windows\system32\svchost.exe -k netsvcs
928 svchost.exe C:\Windows\system32\svchost.exe -k LocalService
1104 spoolsv.exe C:\Windows\system32\spoolsv.exe -k NetworkService
1160 svchost.exe C:\Windows\system32\svchost.exe -k LocalServiceNoNetwork
1236 taskhost.exe "taskhost.exe"
1344 dwm.exe "C:\Windows\system32\Dwm.exe"
1356 explorer.exe C:\Windows\Explorer.EXE
1508 VGAuthService.exe "C:\Program Files\VMware\VMware Tools\VGAuthService.exe"
1576 vmtoolsd.exe "C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" -n vmusr
1604 vmtoolsd.exe "C:\Program Files\VMware\VMware Tools\vmtoolsd.exe"
1648 ManagementAgentHost.exe "C:\Program Files\VMware\VMware Tools\ManagementAgentHost.exe"
1968 svchost.exe C:\Windows\system32\svchost.exe -k bthsvcs
1444 WmiPrvSE.exe C:\Windows\system32\wbem\wmiPrvse.exe
868 dlhhost.exe C:\Windows\system32\dlhhost.exe /Processid:{02D4B3F1-FD88-11D1-960D-00805FC79235}
2064 msdtc.exe C:\Windows\System32\msdtc.exe
2356 SearchIndexer.exe C:\Windows\system32\SearchIndexer.exe /Embedding
2508 wmpnetwk.exe "C:\Program Files\Windows Media Player\wmpnetwk.exe"
2620 svchost.exe C:\Windows\system32\svchost.exe -k LocalServiceAndNoImpersonation
3016 GoogleCrashHandler.exe "C:\Program Files (x86)\Google\Update\1.3.36.292\GoogleCrashHandler.exe"
3028 GoogleCrashHandler.exe "C:\Program Files (x86)\Google\Update\1.3.36.292\GoogleCrashHandler64.exe"
2748 lsass.exe "C:\Users\0XSH3R~1\AppData\Local\Temp\925e7e99c5\lsass.exe"
3064 rundll32.exe "C:\Windows\System32\rundll32.exe" C:\Users\0XSH3R~1\AppData\Roaming\116711e5a2ab05\clip64.dll, Main
2124 taskeng.exe {08C69956-101C-473F-92F2-258443DE915D}
2592 sppsvc.exe C:\Windows\system32\sppsvc.exe
2476 cmd.exe Required memory at 0x7ffffdb020 is not valid (process exited?)
2428 conhost.exe Required memory at 0x7ffffdb020 is not valid (process exited?)
1744 ipconfig.exe Required memory at 0x7ffffdb020 is not valid (process exited?)
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$
```

ANSWER:C:\Users\0XSH3R~1\AppData\Local\Temp\925e7e99c5\lsass.exe

Q3.Persistent external communications suggest the malware's attempts to reach out C2C server. Can you identify the Command and Control (C2C) server IP that the process interacts with?

for this section the command that i use is just to update the volatility into "windows.netscan", but after running that, it gives me a lot of noises, so to filter out what i am looking for, i just use the grep and the unique id of the lsass.exe which "2748"

```
Terminal - ubuntu@ip-172-31-25-79: ~/Desktop/Start here/Tools/volatility3
File Edit View Terminal Tabs Help
0x1d9ac5f0 UDPv4 0.0.0.0 3702 * 0 368 svchost.exe 2023-08-09 21:32:45.000000
0x1daa0720 UDPv4 0.0.0.0 53407 * 0 2620 svchost.exe 2023-08-09 21:32:18.000000
0x1daa0720 UDPv6 :: 53407 * 0 2620 svchost.exe 2023-08-09 21:32:18.000000
0x1db7ea10 TCPv4 0.0.0.0 49154 0.0.0.0 0 LISTENING 924 svchost.exe -
0x1db7ea10 TCPv6 :: 49154 :: 0 LISTENING 924 svchost.exe -
0x1dbb3640 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 716 svchost.exe -
0x1dbb50b0 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 716 svchost.exe -
0x1dbb50b0 TCPv6 :: 135 :: 0 LISTENING 716 svchost.exe -
0x1dbc1010 TCPv4 0.0.0.0 49152 0.0.0.0 0 LISTENING 400 wininit.exe -
0x1dbc14a0 TCPv4 0.0.0.0 49152 0.0.0.0 0 LISTENING 400 wininit.exe -
0x1dbc14a0 TCPv6 :: 49152 :: 0 LISTENING 400 wininit.exe -
0x1e61eec0 UDPv4 192.168.195.136 53404 * 0 2620 svchost.exe 2023-08-09 21:32:17.000000
0x1e621700 UDPv6 fe80::8d94:2fed:6c71:eac4 1900 * 0 2620 svchost.exe 2023-08-09 21:32:17.000000
17.000000
0x1e622270 UDPv4 127.0.0.1 53405 * 0 2620 svchost.exe 2023-08-09 21:32:17.000000
0x1e623ec0 UDPv6 fe80::8d94:2fed:6c71:eac4 53402 * 0 2620 svchost.exe 2023-08-09 21:32:17.000000
17.000000
0x1e673ae0 TCPv4 0.0.0.0 445 0.0.0.0 0 LISTENING 4 System -
0x1e673ae0 TCPv6 :: 445 :: 0 LISTENING 4 System -
0x1e6db500 UDPv4 0.0.0.0 53408 * 0 368 svchost.exe 2023-08-09 21:32:18.000000
0x1e928010 TCPv4 0.0.0.0 49155 0.0.0.0 0 LISTENING 500 services.exe -
0x1e928010 TCPv6 :: 49155 :: 0 LISTENING 500 services.exe -
0x1e94dcf0 TCPv4 192.168.195.136 49168 41.75.84.12 80 CLOSED 2748 lssass.exe N/A
0x1e9846b0 UDPv4 192.168.195.136 68 * 0 768 svchost.exe 2023-08-09 21:50:07.000000
0x1e988840 UDPv4 0.0.0.0 5355 * 0 920 svchost.exe 2023-08-09 21:50:07.000000
0x1e988840 UDPv6 :: 5355 * 0 920 svchost.exe 2023-08-09 21:50:07.000000
0x1ef43830 UDPv6 ::1 53403 * 0 2620 svchost.exe 2023-08-09 21:32:17.000000
0x1f1dd690 TCPv4 0.0.0.0 5357 0.0.0.0 0 LISTENING 4 System -
0x1f1dd690 TCPv6 :: 5357 :: 0 LISTENING 4 System -
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$ vol.py -q -f '/home/ubuntu/Desktop/Start here/Artifacts/Windows 7
x64-Snapshot4.vmem' windows.netscan | grep 2748
0xe239910 TCPv4 - 0 56.75.178.2 0 CLOSED 2748 lssass.exe N/A
0x1d75b530 TCPv4 192.168.195.136 49167 41.75.84.12 80 CLOSED 2748 lssass.exe N/A
0x1e94dcf0 TCPv4 192.168.195.136 49168 41.75.84.12 80 CLOSED 2748 lssass.exe N/A
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$
```

ANSWER: 41.75.84.12

Q4. Following the malware link with the C2C, the malware is likely fetching additional tools or modules. How many distinct files is it trying to bring onto the compromised workstation?

for this question made me confused, but thanks for the help on the community from the internet. made some research on how to get the the distinct files, so basically i need to dump first the pid using this command:

```
vol.py -f ~/Desktop/Start\ here/Artifacts/Windows\ 7\ x64-Snapshot4.vmem windows.memmap -
-pid 2748 --dump
```

after that command i follow it with this:

```
strings pid.2748.dmp | grep "GET /"
```

that's how i got the answer

```
Terminal - ubuntu@ip-172-31-25-79: ~
File Edit View Terminal Tabs Help

Terminal - ubuntu@ip-172-31-25-79: ~/Desktop/Start here/Tools/volatility3
File Edit View Terminal Tabs Help

0xfa800fdfe000 0xd9d000 0x1000 0xa5c8000 pid.2748.dmp
0xfa800fedb000 0xc2e000 0x1000 0xa5c9000 pid.2748.dmp
0xfa8010028000 0xd7c000 0x1000 0xa5ca000 pid.2748.dmp
0xfa801014b000 0x345f000 0x1000 0xa5cb000 pid.2748.dmp
0xfa80101e8000 0x7bc000 0x1000 0xa5cc000 pid.2748.dmp
0xfa8010324000 0x1c8f9000 0x1000 0xa5cd000 pid.2748.dmp
0xfa80103b9000 0x350e000 0x1000 0xa5ce000 pid.2748.dmp
0xfa80103f9000 0x1d369000 0x1000 0xa5cf000 pid.2748.dmp
0xfa80103ff000 0x1d1ef000 0x1000 0xa5d0000 pid.2748.dmp
0xfa80105ba000 0x1d3d0000 0x1000 0xa5d1000 pid.2748.dmp
0xfa80105fe000 0x1cf03000 0x1000 0xa5d2000 pid.2748.dmp
0xfa80107c7000 0x1cdd000 0x1000 0xa5d3000 pid.2748.dmp
0xfa80107ff000 0x1cde8000 0x1000 0xa5d4000 pid.2748.dmp
0xfa8010908000 0x1cce0000 0x1000 0xa5d5000 pid.2748.dmp
0xfa80109b3000 0x1d04b000 0x1000 0xa5d6000 pid.2748.dmp
0xfa80109fa000 0x1c155000 0x1000 0xa5d7000 pid.2748.dmp
0xfa80109fb000 0x1ce56000 0x1000 0xa5d8000 pid.2748.dmp
0xfa80109fc000 0x1ce17000 0x1000 0xa5d9000 pid.2748.dmp
0xfa8010a89000 0x1cce2000 0x1000 0xa5da000 pid.2748.dmp
0xfa8010aab000 0x1cb04000 0x1000 0xa5db000 pid.2748.dmp
0xfa8010b6b000 0x1ca44000 0x1000 0xa5dc000 pid.2748.dmp
0xfa8010bf9000 0x1cd53000 0x1000 0xa5dd000 pid.2748.dmp
0xffffffff0000 0x100000 0x1000 0xa5de000 pid.2748.dmp
0xffffffff0100 0x105000 0x1000 0xa5df000 pid.2748.dmp
0xffffffff0200 0x101000 0x4000 0xa5e0000 pid.2748.dmp
0xffffffff0700 0x3000 0x1000 0xa5e4000 pid.2748.dmp
0xffffffff0800 0x1000 0x2000 0xa5e5000 pid.2748.dmp
0xffffffff0b00 0x4000 0x3000 0xa5e7000 pid.2748.dmp
0xffffffff0e00 0x0 0x1000 0xa5ea000 pid.2748.dmp
0xffffffff0f00 0x7000 0x2000 0xa5eb000 pid.2748.dmp
0xffffffff2800 0x40000 0x1f000 0xa5ed000 pid.2748.dmp

ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$ strings pid.2748.dmp | grep "GET /"
GET /rock/Plugins/cred64.dll HTTP/1.1
GET /rock/Plugins/clip64.dll HTTP/1.1
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$
```

ANSWER: 2

Q5. Identifying the storage points of these additional components is critical for containment and cleanup. What is the full path of the file downloaded and used by the malware in its malicious activity?

this question is not really complicated for me, the question is just looking for the full path of the downloaded file that used by the malware. so here is the command that i used to find it

```
Terminal - ubuntu@ip-172-31-25-79: ~
File Edit View Terminal Tabs Help

Terminal - ubuntu@ip-172-31-25-79: ~/Desktop/Start here/Tools/volatility3
File Edit View Terminal Tabs Help

ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$ vol.py -q -f '/home/ubuntu/Desktop/Start here/Artifacts/Windows 7
x64-Snapshot4.vmem' windows.cmdline | grep dll
868      dllhost.exe      C:\Windows\system32\dllhost.exe /Processid:{02D4B3F1-FD88-11D1-960D-00805FC79235}
3064      rundll32.exe      "C:\Windows\System32\rundll32.exe" C:\Users\0xSh3r10ck\AppData\Roaming\116711e5a2ab05\clip64.dll, Main
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$
```

ANSWER: C:\Users\0xSh3r10ck\AppData\Roaming\116711e5a2ab05\clip64.dll

Q6. Once retrieved, the malware aims to activate its additional components. Which child process is initiated by the malware to execute these files?

it is really on the command output that i run from the previous question which is the rundll32.exe

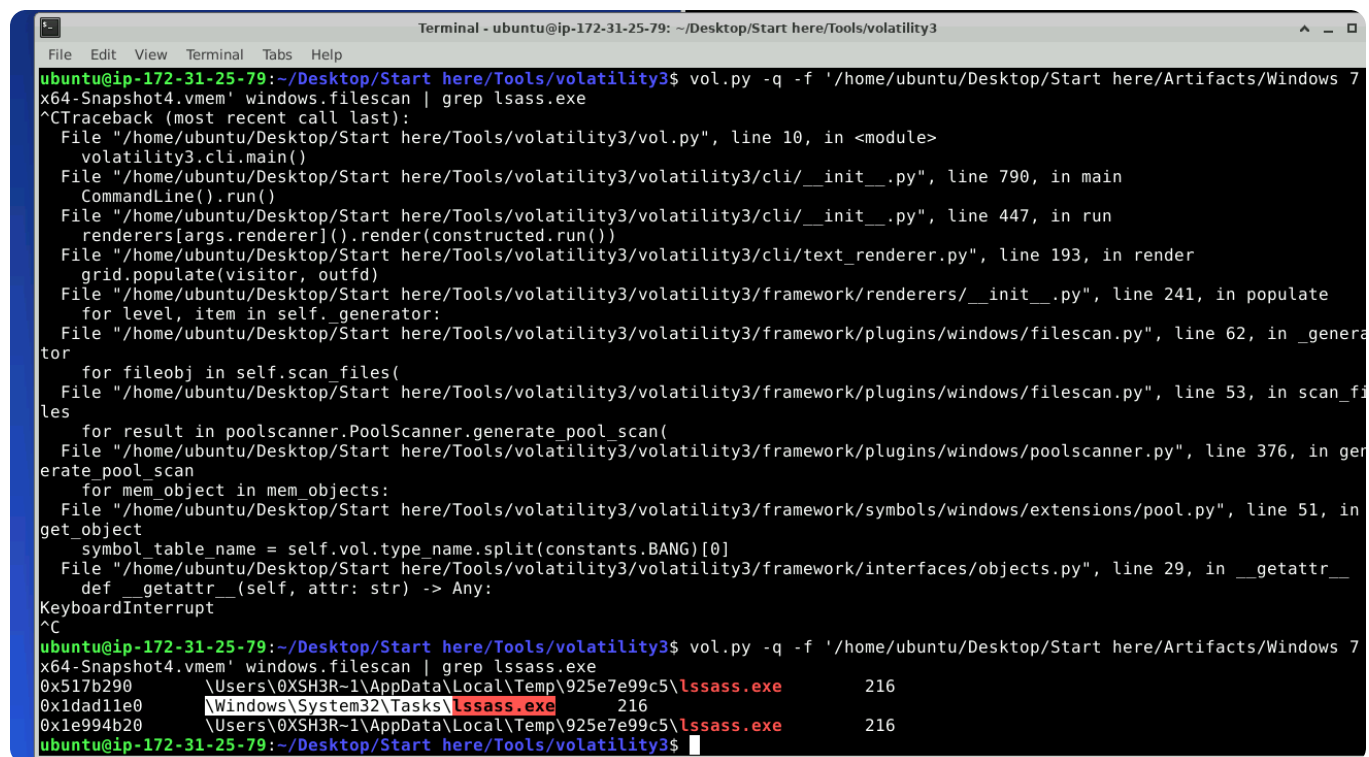
ANSWER: rundll32.exe

Q7. Understanding the full range of Amadey's persistence mechanisms can help in an effective mitigation. Apart from the locations already spotlighted, where else might the malware be ensuring its consistent presence?

now, for the last question basically we need to use the filescan from volatility and grep the fake windows process and a malware that we found earlier which is lssass.exe

here is the command that i use:

```
vol.py -q -f '/home/ubuntu/Desktop/Start here/Artifacts/Windows 7 x64-Snapshot4.vmem' windows.filescan | grep lssass.exe
```



```
Terminal - ubuntu@ip-172-31-25-79: ~/Desktop/Start here/Tools/volatility3
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$ vol.py -q -f '/home/ubuntu/Desktop/Start here/Artifacts/Windows 7
x64-Snapshot4.vmem' windows.filescan | grep lssass.exe
^C
Traceback (most recent call last):
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/vol.py", line 10, in <module>
    volatility3.cli.main()
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/cli/_init_.py", line 790, in main
    CommandLine().run()
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/cli/_init_.py", line 447, in run
    renderers[args.renderer]().render(constructed.run())
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/cli/text_renderer.py", line 193, in render
    grid.populate(visitor, outfd)
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/framework/renderers/_init_.py", line 241, in populate
    for level, item in self.generator:
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/framework/plugins/windows/filescan.py", line 62, in _genera
tor
    for fileobj in self.scan_files():
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/framework/plugins/windows/filescan.py", line 53, in scan_fi
les
    for result in poolscanner.PoolScanner.generate_pool_scan():
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/framework/plugins/windows/poolscanner.py", line 376, in ge
nerate_pool_scan
    for mem_object in mem_objects:
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/framework/symbols/windows/extensions/pool.py", line 51, in
get_object
    symbol_table_name = self.vol.type.name.split(constants.BANG)[0]
  File "/home/ubuntu/Desktop/Start here/Tools/volatility3/volatility3/framework/interfaces/objects.py", line 29, in __getattr__
    def __getattr__(self, attr: str) -> Any:
KeyboardInterrupt
^C
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$ vol.py -q -f '/home/ubuntu/Desktop/Start here/Artifacts/Windows 7
x64-Snapshot4.vmem' windows.filescan | grep lssass.exe
0x517b290      \Users\0XSH3R-1\AppData\Local\Temp\925e7e99c5\lssass.exe      216
0x1dad11e0    \Windows\System32\Tasks\lssass.exe      216
0x1e994b20    \Users\0XSH3R-1\AppData\Local\Temp\925e7e99c5\lssass.exe      216
ubuntu@ip-172-31-25-79:~/Desktop/Start here/Tools/volatility3$
```